Margaret Giardello

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Professional Summary

Recent Mater's graduate and emerging data scientist with hands-on research experience spanning experimental design, statistical analysis, data engineering, and visualization (R, Python, SQL). Skilled at translating complex, multi-source data into clear, actionable findings for audiences ranging from academic conferences to public-facing exhibits to nonprofit stakeholders. Earned Master's degree with a focus on interdisciplinary applications and building end-to-end data pipelines that turn raw data into decision-ready insight.

Areas of Expertise

- R
- Python
- SQL
- Shiny Applications
- Grafana Dashboarding
- Data Engineering
- Data Visualization
- Github Integration
- Statistical Analysis
- Machine Learning
- API Integration
- Research Design
- Qualtrics
- Technical Writing
- B1 Spanish

Relevant Professional Experience

Research Assistant, Lewis & Clark College | *Portland, OR* **Sept 2024 – Sept 2025**

- As a promotion from the Rogers Fellowship program, continued research design and execution
- Took leadership of new research aides by creating reference documentation, becoming the primary contact for questions, overseeing design implementation, and compiling team results
- Recruited participants and conducted 100+ experimental procedure sessions
- Independently created a system of data collection, organization, and storage using Qualtrics and connected Google platforms to streamline data processing and ensure zero loss of data
- Acted as the sole research assistant in data cleaning and analysis using google sheets, excel, SPSS, and R to prepare findings for formal presentation at academic and emergency preparedness conferences
- Represented the project at OMSI After Dark, guiding 500+ attendees through a simplified version of the study protocol (https://www.cascadia9game.org/)

Data Analyst, Growing Gardens**,** Lewis & Clark College | *Portland, OR* **Jan – May 2025**

- Worked in a small team to complete analysis tasks related to participation in the school program of Growing Gardens, a local Portland non-profit serving 10 schools in food security events
- Utilized R for data cleaning, aggregation, analysis, and visualization
- Developed survey materials using google forms and formal documentation of organization and storage
- Created a report of preliminary results and recommendations for future data gathering processes to lessen gathering burden on volunteer teachers and ensure future data validity
- Work featured in an OPB radio segment and article, extending the project's visibility beyond the college radio segment and article

Student Data Science Assistant, Lewis & Clark College | *Portland, OR* **Sept 2024– May 2025**

- Supported 30+ students per term as TA for introductory data science courses, resolving in-lecture technical issues and grading assignments
- Staffed the campus 'Data Cafe' as an R and data-sourcing specialist
- Improved student outcomes through office hours, 1:1 tutoring, and cross-department faculty referrals

Rogers Fellowship Research Lewis & Clark College | *Portland, OR* **May - Aug 2024**

- Designed and executed Phase III of an NSF-funded (#1917148) interdisciplinary study on video games for adolescent disaster preparedness
- Conducted a literature review and formal experimental proposal that built directly on prior study limitations and reflected constraints in data collection

- Personally recovered and re-analyzed a backlog of unused footage/data by building a new annotation and storage system, salvaging results of 100+ experimental sessions
- Coordinated with participant pools, fellow research aides, and PIs to refine game materials for testing
- Conducted basic statistical analysis of data cross experiments using google sheets, excel, SPSS, and R to assess experimental results and guide future experiment design
- Documented and presented research in a formal paper and academic presentation

Data Analyst, US Fish & Wildlife Service, Lewis & Clark College | *Portland, OR* **Jan – May 2024**

- Built a species distribution assessment predicting future habitable range for the fisher (Pekania pennanti) under shifting climate conditions
- Used R and GitHub for a full data pipeline: cleaning, aggregation, analysis, and visualization
- Produced a paper and presentation for conservation stakeholders at the US Fish & Wildlife Service (https://zenodo.org/records/11051172)

Projects

Media Indicators of Misinformation | Python, R, SQL, Github, Machine Learning **May 2026 - Aug 2026**

- Built a capstone pipeline aggregating article-level text analysis across global news sources as predictors of country-year misinformation campaigns
- Engineered a layered SQL view over the GDELT database in Google BigQuery to support the transition from 2.6TB of data to ~3000 rows for aggregated analysis
- Cleaned, filtered, joined, and transformed data in a combination of R, Python, & SQL languages
- Applied Random Forest modeling, PCA, and clustering for dimensionality reduction
- Developing a multivariate time series model (Random Forest, XGBoost) to predict the presence of domestic misinformation campaigns on a global scale
- Reported results via formal report and conference poster

Prague Metro System | SQL, Grafana, DBT Modeling **May 2026 - Aug 2026**

- Built dimensional models on a synthetic transit data form a variety of datasource structures to represent metro system usage patterns, as part of a 3-person team
- Engineered Python DAGs for ingestion, dbt/SQL for modeling, and Postgres for storage
- Delivered biweekly Grafana dashboards tailored to specific business questions, enabling theoretical transit design tailored to user movement patterns

Portland Lighting & Crime | R, ShinyApps, Statistical Analysis, Data Visualization **Jan - May 2026**

- Analyzed 2025 Portland crime and streetlight data to test whether artificial lighting deters targeted crime categories relevant to pedestrians traveling at night
- Cleaned and spatially joined and transformed disparate open-data sources in R
- Ran density analysis and chi-square testing, supporting working theory and calling for more tailored future analysis which accounts for proxy variable relationships and temporal analysis
- Built an interactive Shiny app letting users explore results by neighborhood and crime type

Yelp Cuisines | YelpAPI, PostgreSQL, Railway, Beekeeper, Grafana **Jan - May 2026**

- Compared restaurant/cuisine patterns across 6 major US cities to surface consumer behavior insights
- Built a data pipeline from the Yelp Business API through Railway into Postgres, visualized in Grafana
- Found preliminary results of higher food ratings and lower costs along West Coast cities, useful for informing future pipeline automation and work in consumer behavior

Education

Master of Science, Willamette University | *Portland, OR* **Aug 2026**
Data Science GPA: 4.0

Bachelor of Arts, Lewis & Clark College | *Portland, OR* **May 2025**
Major: Psychology Minor: Data Science GPA: 3.80